Problem 3

a) Markov Inequality

$X = X \cdot 1_{\{X \leq r\}} + X \cdot 1_{\{X > r\}}$ 1 being the indicator function

$E[X] = E[X \cdot 1_{\{X \leq r\}} + X \cdot 1_{\{X > r\}}] = E[X \cdot 1_{\{X \leq r\}}] + E[X \cdot 1_{\{X > r\}}]$

$\geq E[X \cdot 1_{\{X > r\}}]$ (X positive)

$\geq E[r \cdot 1_{\{X > r\}}] = r\, E[1_{\{X > r\}}] = r \cdot P_r\{X > r\}$

$\Rightarrow P_r\{X > r\} \leq E[X]/r$ (Similar idea can be done without 1_E explicitly, using the sum or integral)

b) Chebychev inequality

Let's set $X = |Y - m|$

with X such that $P(X=0)=1$

$P(X > r) = 0 = \frac{E[X]}{r}$

we have $P_r\{|X-m| > \varepsilon\} = P_r\{(X-m)^2 > \varepsilon^2\} = P_r\{Y > \varepsilon^2\}$

Using Markov $\Rightarrow P_r\{Y > \varepsilon^2\} \leq \frac{E[Y]}{\varepsilon^2}$

$E[Y] = E[(X-m)^2] = Var(X)$

Finally $\Rightarrow P_r(|X-m| > \varepsilon) \leq \frac{Var(X)}{\varepsilon^2}$

c) Weak Law of Large Numbers

Using Chebychev with $\bar{Z}_n = \frac{1}{n}\sum_{i=1}^{n} Z_i$

we get $P_r\{|\bar{Z}_n - m| > \varepsilon\} \leq \frac{Var(\bar{Z}_n)}{\varepsilon^2}$

$Var(Z_n) = Var\left(\frac{1}{n}\sum_{i=1}^{n} Z_i\right) = \frac{1}{n^2} Var\left(\sum_{i=1}^{n} Z_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} Var(Z_i)$

$= \frac{n}{n^2} Var(Z) = \frac{1}{n} Var(Z)$ independents

Finally $\Rightarrow P_r\{|\bar{Z}_n - m| > \varepsilon\} \leq \frac{Var(Z)}{n\,\varepsilon^2}$